

The human brain can imagine, think, and compute amazingly well, and only consumes 500 calories a day. Why are we convinced that AI requires vast amounts of energy and increasingly expensive datacenter usage?

Posted by u/achicomp • 0 points • 283 comments

Why is the assumption that today and in the future we will need ridiculous amounts of energy expenditure to power very expensive hardware and datacenters costing billions of dollars, when we know that a human brain is capable of actual general intelligence at very small energy costs? Isn't the human brain an obvious real life example that our current approach to artificial intelligence is not anywhere close to being optimized and efficient?

Comments

u/smswigart • 1 points • 2025-06-26 18:22 UTC

AI will continue to use fewer Joules per Token. It's unknown what the lower limit is, but our brain is probably pretty close. Look up Brain Organoid AI if you want some nightmare fuel.

u/data_snacks • 1 points • 2025-06-26 15:10 UTC

This is a bit scary: [<https://www.visualcapitalist.com/charted-productivity-gains-from-using-ai/>] (<https://www.visualcapitalist.com/charted-productivity-gains-from-using-ai/>) \- how much time we gain by using generative AI in different tasks

u/DurtyKurty • 1 points • 2025-06-25 20:15 UTC

I have to use calories to get my brain to a school to make my brain better. I have to use calories in all the between times my brain is doing jack shit. I have to entertain my brain for many hours to convince it to learn something new afterwards. Some people's brain and body use the same amount of calories but are completely sub par. Our brains have thought up a hundred thousand ways to kill other different brains.

I'm not discounting how energy efficient our bodies are. They're pretty good at doing a million things with nothing but tacos for gas.

u/The_Justice_Man • 1 points • 2025-06-25 13:50 UTC

Currently AI uses minimal energy. Way less than gaming on a PC. It might be a few times more energy than your smartphone uses(per person).

I have no worry that it'll ever use much more power than it does now.

u/RangerRealistic200 • 1 points • 2025-06-25 05:25 UTC

THERE'S A NEED FOR BETTER SYSTEMS IN AI

u/RangerRealistic200 • 1 points • 2025-06-25 05:24 UTC

The Brain takes only about 12 watts to function with a 100 billion neurons that would take lifetimes for AI

u/[deleted] • 1 points • 2025-06-24 22:55 UTC

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u/Isogash • 1 points • 2025-06-24 21:46 UTC

You're exactly right, we shouldn't assume.

However, the AI industry right now is mostly built on hype that companies are looking to deliver on ASAP to keep investments up, and the only reliable way to achieve better results right now is with increased computing power. We don't know what we're missing until we find it.

There's a very famous brief article called The Bitter Lesson which has been very influential, you can look it up.

u/smokefoot8 • 1 points • 2025-06-24 16:19 UTC

You can run DeepSeek on a personal computer, so vast amounts of compute and energy aren't required - for a single instance of the AI model. But companies don't want just one copy of the AI answering questions and faking photos - they want thousands!

The number of AI agents will increase with no limit except how much money companies are willing to spend. Since they are vastly cheaper than the employees they replace, companies will be willing to spend a lot on data centers and the energy to run them.

So don't compare AI demand with a single human brain, but with millions of human brains.

u/_orbital • 1 points • 2025-06-24 15:47 UTC

because we haven't even come close to cracking the code of how the brain does 17 dimensional quantum computation...

u/Conscious_Cut_6144 • 1 points • 2025-06-24 14:00 UTC

500 calories of cheeseburger,
That cow ate 50000 calories of grass for that meat
And that grass needed 5,000,000 calories of sunlight.

Aka 5800KWh

After solar panels efficiency that's about 1000KWh of usable electricity.

That's enough electricity to run a rack full of ChatGPT servers, enough to serve hundreds of users for 24 hours.

We aren't that efficient.

u/Secure_Candidate_221 • 1 points • 2025-06-24 13:39 UTC

Simple mate. The human brain is different from AI

u/Minimum_Minimum4577 • 1 points • 2025-06-24 10:10 UTC

Facts. The human brain runs on a sandwich and a nap, while AI needs a warehouse and a power plant. Definitely shows we're still figuring it out.